

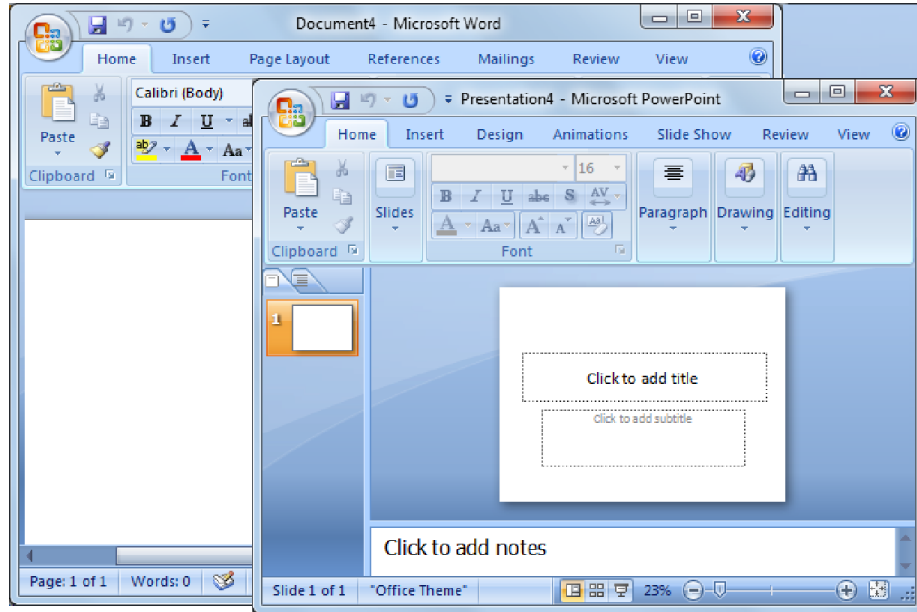
Golden Rules and Heuristics

Slides by Kiran Khurshid

Shneiderman's Eight Golden Rules of Interface Design

1. Strive for consistency.
2. Seek universal usability (*Previously stated as, and still renowned as: Enable frequent users to use shortcuts*).
3. Offer informative feedback.
4. Design dialogs to yield closure.
5. Prevent errors.
6. Permit easy reversal of actions.
7. Keep users in control.
8. Reduce short-term memory load

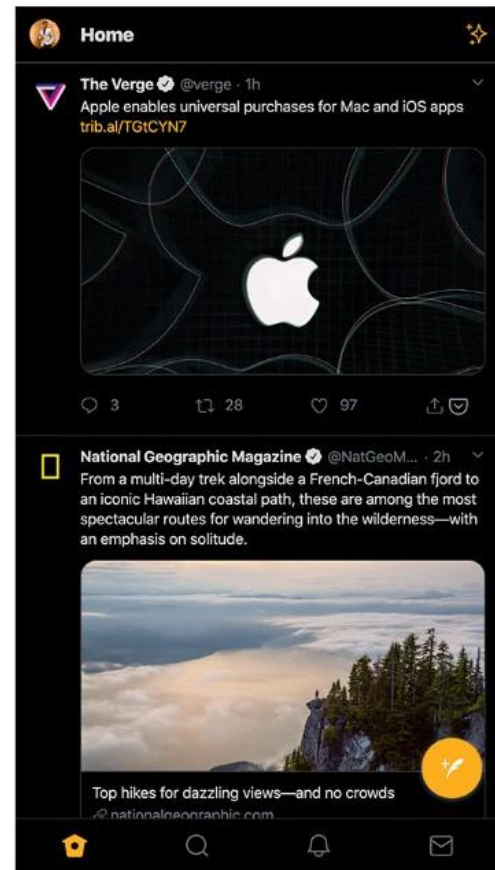
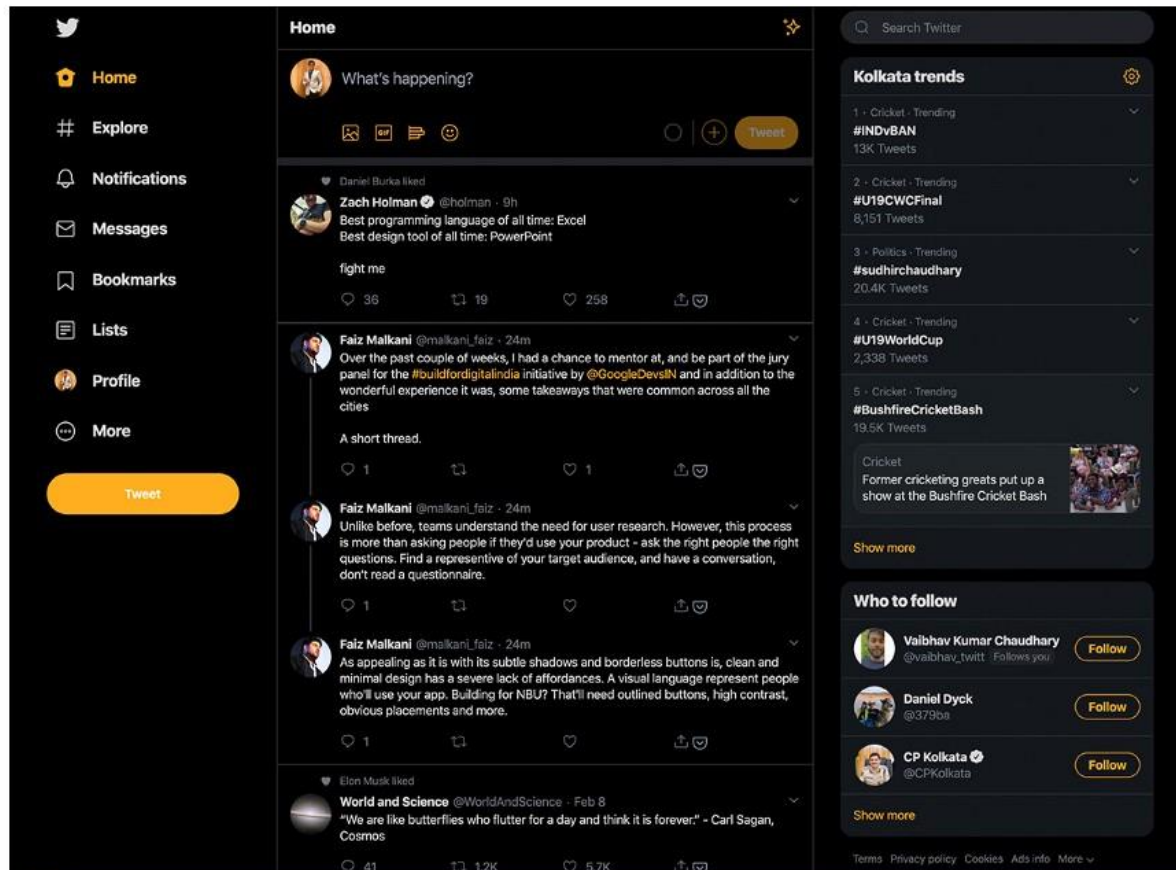
1. Strive for consistency



Same shortcut keys for same actions



Consistency (cont'd)



2. Seek universal usability (or enable frequent users to use shortcuts)



Keyboard shortcuts speed up workflows for frequent users

2. (Cont'd)

Photoshop offers shortcuts for frequent users

Ps CS6 Shortcuts

										Brushes Palette F5	Color Palette F6	Layer Palette F7	Info Palette F8	Actions Palette F9	Revert F12
Switch Open Tabs ~	10% Opacity 1	20% Opacity 2	30% Opacity 3	40% Opacity 4	50% Opacity 5	60% Opacity 6	70% Opacity 7	80% Opacity 8	90% Opacity 9	100% Opacity 0 Screen	Fit -	Zoom In =	Zoom Out +	Backspace	
Tab	Quick Mask Q	Magic Wand W	Eraser E	Rotate Tool R	Type Tool T	History Brush Y	Shape Tool U	Eye Dropper I	Dodge Tool O	Pen Tool P	{ Brush Size [Move Layer ↓	+ Brush Size] Move Layer ↑	Toggle Layer Mask \		
Caps Lock	Path Select A	Clone Stamp S	Default Colors D	Screen Modes F	Paint Bucket G	Hand H	Healing Brush J	3D Tools K	Lasso Tools L	: Hide/Show Guides "	" Hide/Show Grid '	Enter			
Shift	Zoom Tool Z	Switch Fg/Bg X	Crop Tool C	Move V	Brush Tool B	3D Camera New File N	Marquee Tools M	Preset Brush 1 ,	Preset Brush 2 .	? /	Shift				
Ctrl	Win	Alt	Space					Alt	Menu	Ctrl					

SWITCHING TOOLS

To switch between tools within groups **Shift** + Tool

SELECTING

Draw Marquee from Center **Alt** + Marquee

Add to a Selection **Shift**

Subtract from a Selection **Alt**

Intersection with a Selection **Shift** + **Alt**

Make Copy of Selection w/ Move Tool **Alt** + Drag Selection

Make Copy of Selection not in Move Tool **Ctrl** + **Alt** + Drag Selection

Move Selection (in 1-pixel increments) **Arrow Keys**

Move Selection (in 10-pixel increments) **Shift** + **Arrow Keys**

VIEWING

Select all Opaque Pixels on Layer **Ctrl** + Click Layer Thumbnail (in Layers Panel)

Restore last Selection **Ctrl** + **Shift** + **D**

Feather Selection **Shift** + **F6**

Move Marquee while drawing Selection Hold Space while drawing marquee

Hide all tools and panels **Tab**

Hide all panels except Toolbox and Options **Shift** + **Tab**

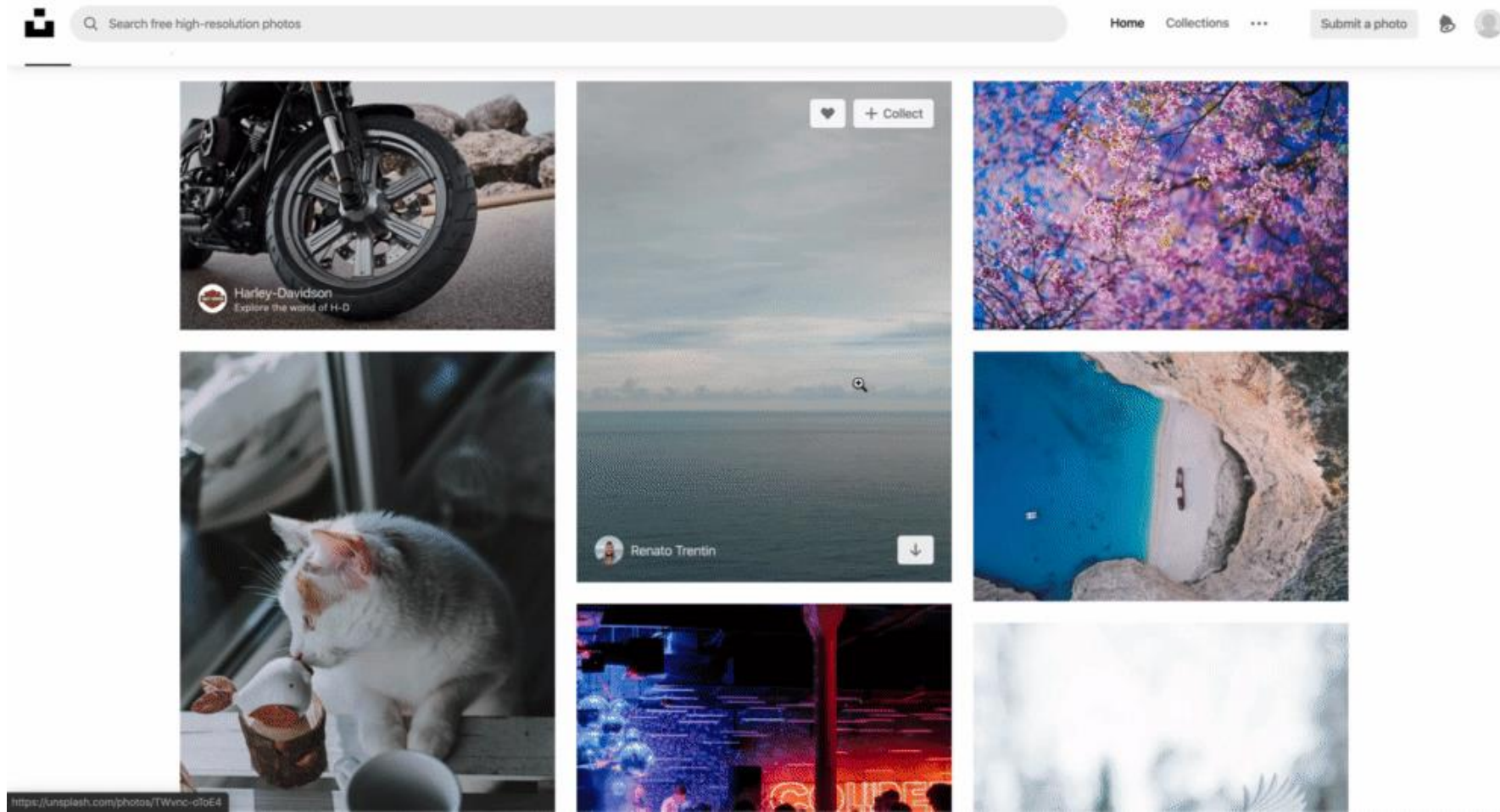
Scroll image left or right in window **Ctrl** + **Shift** + Page / |

Jump/Zoom to part of Image **Ctrl** + **Shift** + Drag in Navigator Panel

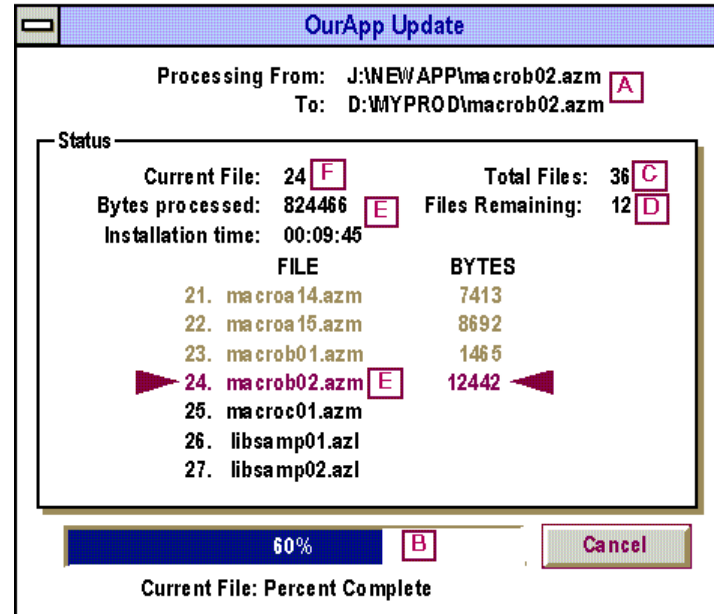
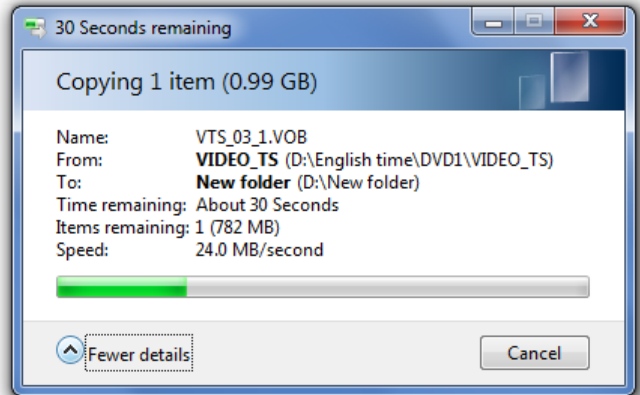
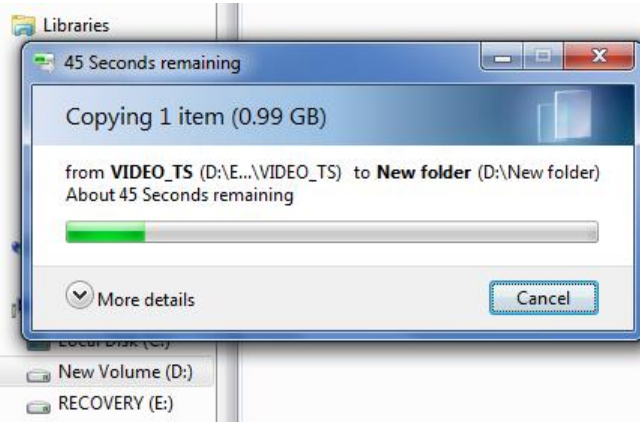
Fit on screen **Ctrl** + **0** or Double-click on Hand Tool

100% View Level (Actual Pixels) **Ctrl** + **Alt** + **0** or Double-click on Zoom Tool

3. Offer informative feedback

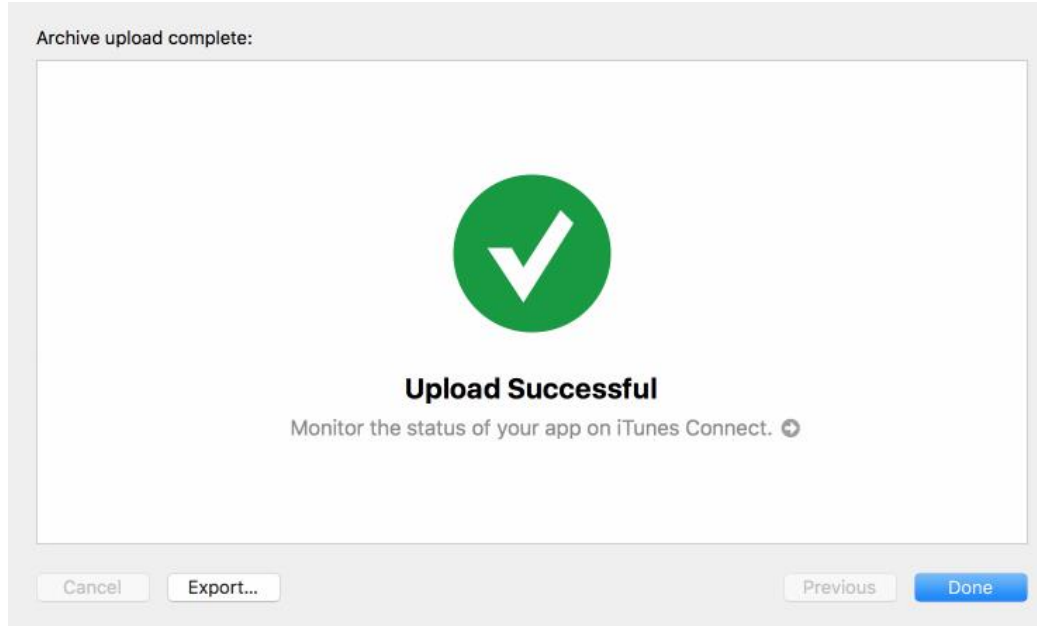


3. Offer informative feedback (Cont'd)

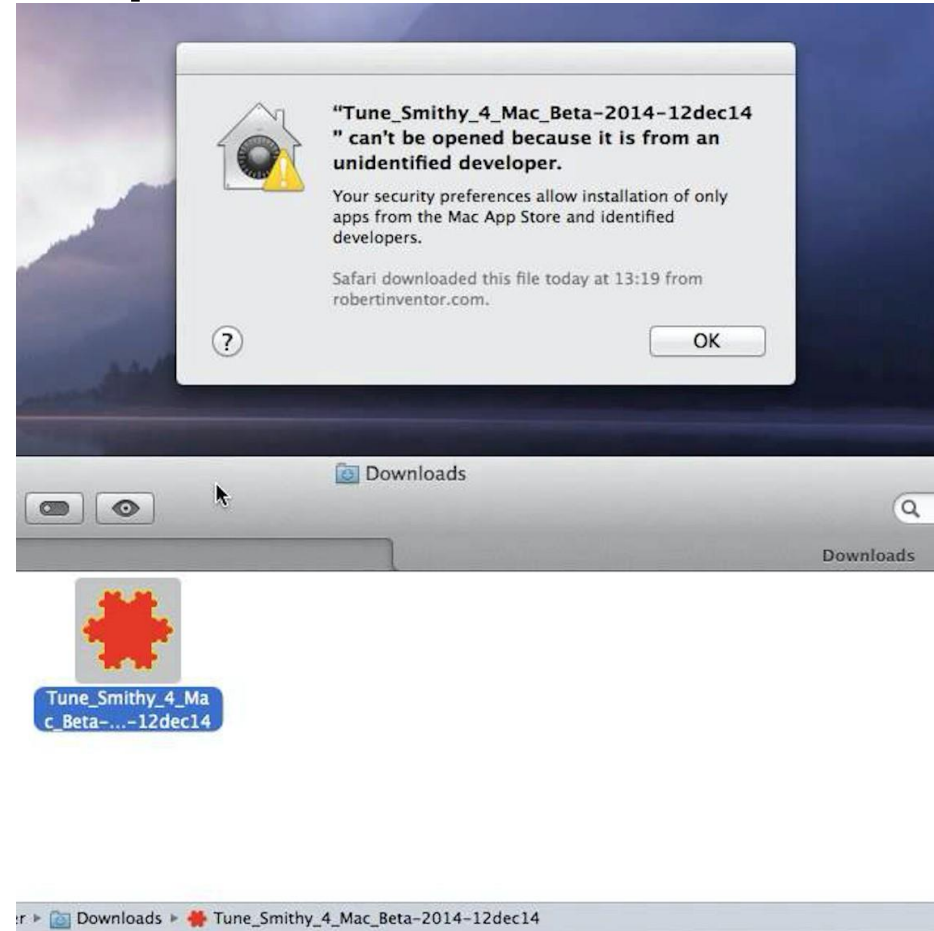
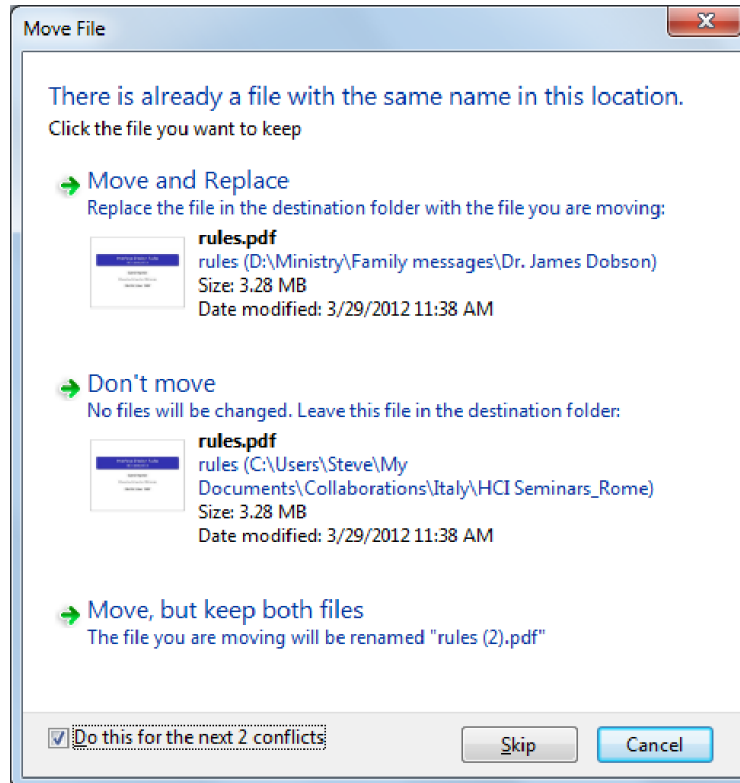


If delay is inevitable, provide reassurance

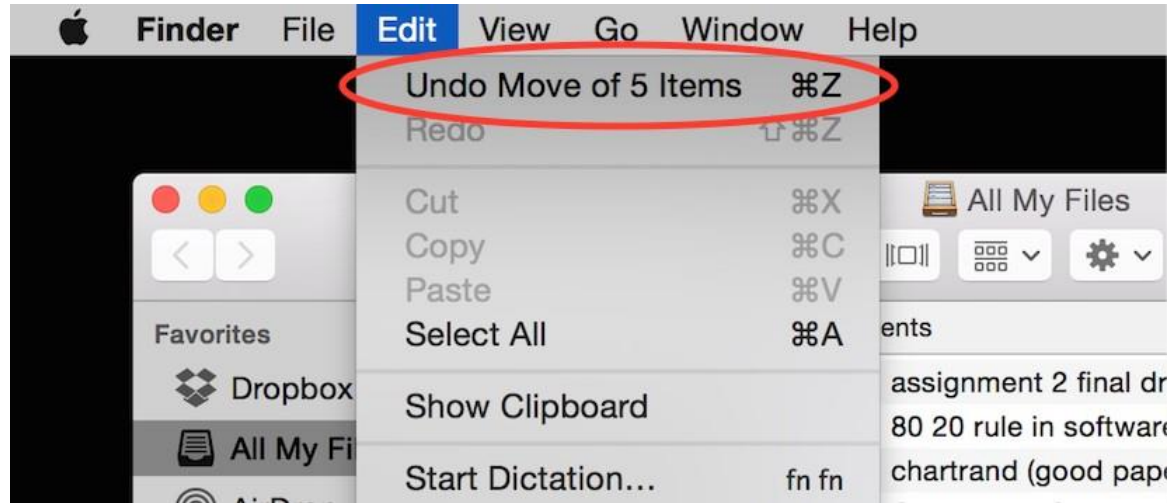
4. Design dialogs to yield closure



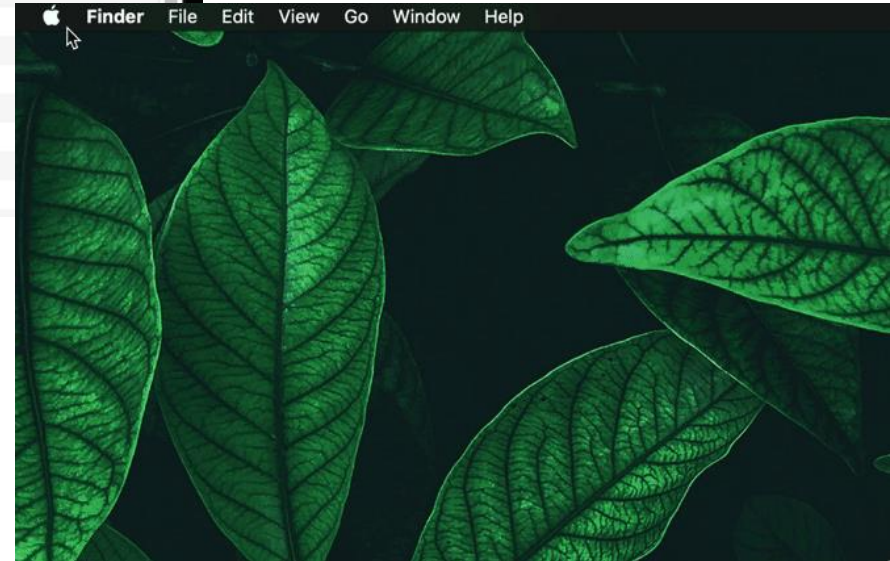
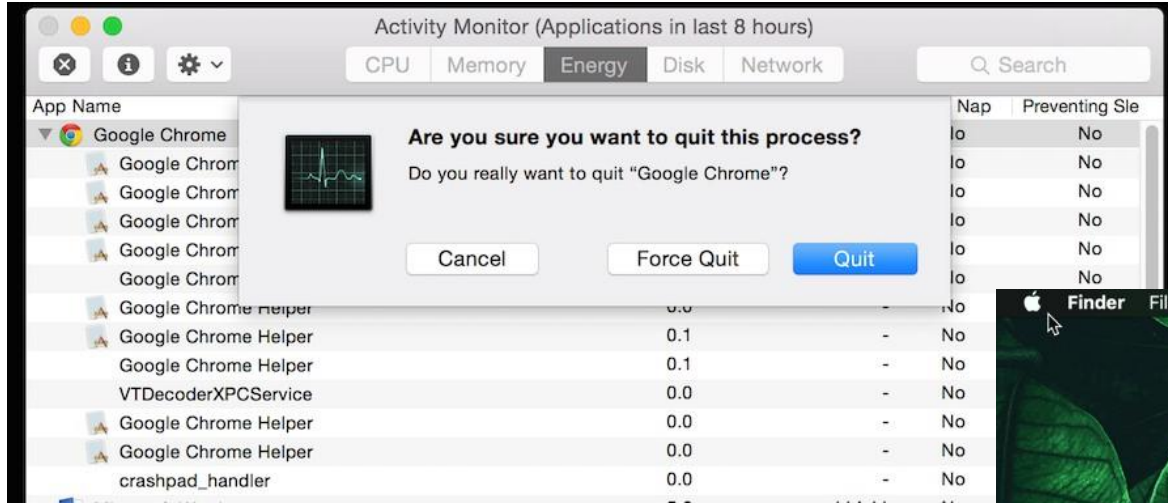
5. Prevent errors (or offer error prevention and simple error handling)



6. Permit easy reversal of actions



7. Keep users in control (or Support internal locus of control)



8. Reduce short term memory load

shows path through web site hierarchy

web site

top level category

sub-category

this page

meandeviation.com > statistics tutorial > notes > 7 errors

7 common errors

live links to higher levels

Some common errors arise again and again in statistics.
Here are some to watch out for.

amazon.in
ben shneiderman
Hello, Rounak
Account & Lists
Returns & Orders
Try Prime
Cart

Deliver to ROUNAK
Kolkata 700012

Buy Again Rounak's Amazon.in Today's Deals

AMAZON APP JACKPOT

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Designing the User Interface: Strategies for Effective Human-Computer Interaction: United States Edition Hardcover – Import, 15 Jul 1997
by Ben Shneiderman (Author)
★★★★☆ 3 ratings

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There is a newer edition of this item:
Designing the User Interface: Strategies for Effective Human-Computer Interaction

In revising this well-known book, Ben Shneiderman again provides a complete, current, and authoritative introduction to user-interface design. Students will learn practical techniques and guidelines needed to develop good systems designs - systems with interfaces the typical user can understand, predict, and control. This third edition...

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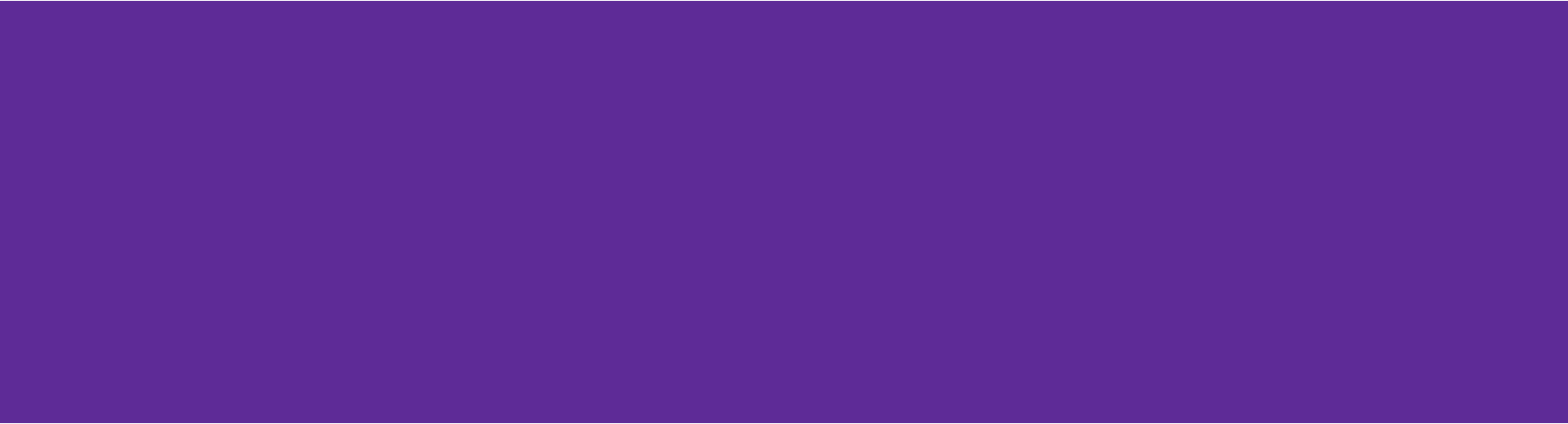
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Shneiderman's 8 Golden Rules of Interface Design

The principles	Questions to consider	Mark Complete
1. Strive for consistency	Is the style of this element maintained across your site/app? Is this content placed in the correct location according to the site hierarchy? Does this follow the conventions for your chosen platform? How can you make your designs more consistent?	☐
2. Enable frequent users to use shortcuts	Are there shortcuts available for your more experienced users? Who is this product designed for? Will there be a need to consider experienced users? How can you make it easier and quicker for experienced users?	☐
3. Offer informative feedback	Does the user know where they are at in the process? Does the user know what they have done after performing this action? How are you communicating this feedback to your user?	☐
4. Design dialogue to yield closure	Does the user have to do any guessing here? Is it clear and obvious enough for your intended audience? Are there any next steps for the user? How are you communicating the system status with the user?	☐
5. Offer simple error handling	Have you done everything imaginable to prevent this error from happening on your end? Is this error avoidable in the first place? If the user does make an error, how easy is it for them to fix it?	☐
6. Permit easy reversal of actions	How many steps does the user have to take to reverse their actions? Will the user quickly realize they need to reverse the action in the first place? How can you make your users detect the possibility of reversal?	☐
7. Support internal locus of control	Will the user feel in control at this specific touch point in your app? Will they be surprised in an unpleasant manner? Does the site feel easily navigable? Does the user feel safe and in control? How can you make the user feel more safe and in control?	☐
8. Reduce short-term memory load	Are there enough visual cues here for the user to find the functionality or item? Do they have to remember things to understand what's going on? How can you help the user recall?	☐

Norman's Seven Principles for Transforming Difficult Tasks into Simple Ones

Originally taken from “Design of Everyday Things” (aka DOET) by Donald Norman

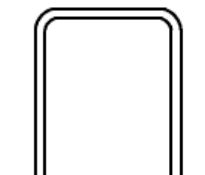


Norman's seven principles

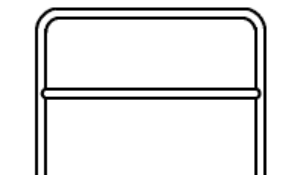
1. Use both knowledge in the world and knowledge in the head
2. Simplify the structure of tasks
3. Make things visible: bridge the gulfs of execution and evaluation
4. Get the mappings right
5. Exploit the power of constraints
6. Design for error
7. When all else fails, standardize

Use both knowledge in the world and knowledge in the head

Knowledge in the head is memory, while knowledge in the world is everything else — external information



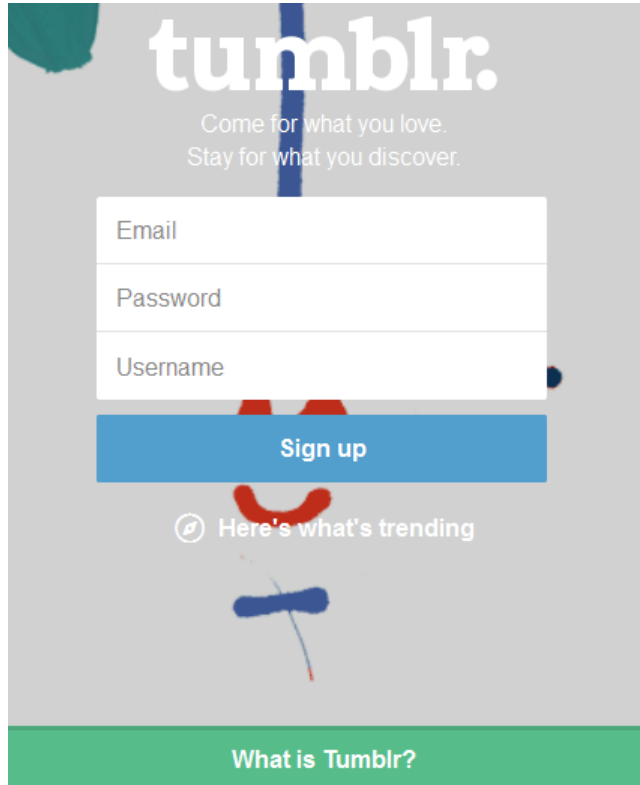
Yellow Springs Rack



Missoula Rack



Simplify the structure of tasks

The image shows a Tumblr sign-up form overlaid on a background featuring the Tumblr logo and abstract brush strokes. The form is a white rectangle with three input fields: 'Email', 'Password', and 'Username'. Below these fields is a blue 'Sign up' button. Under the button is a link that says 'Here's what's trending' with a circular icon containing a 'p'. At the bottom of the form area is a green bar with the text 'What is Tumblr?'.

tumblr.

Come for what you love.
Stay for what you discover.

Email

Password

Username

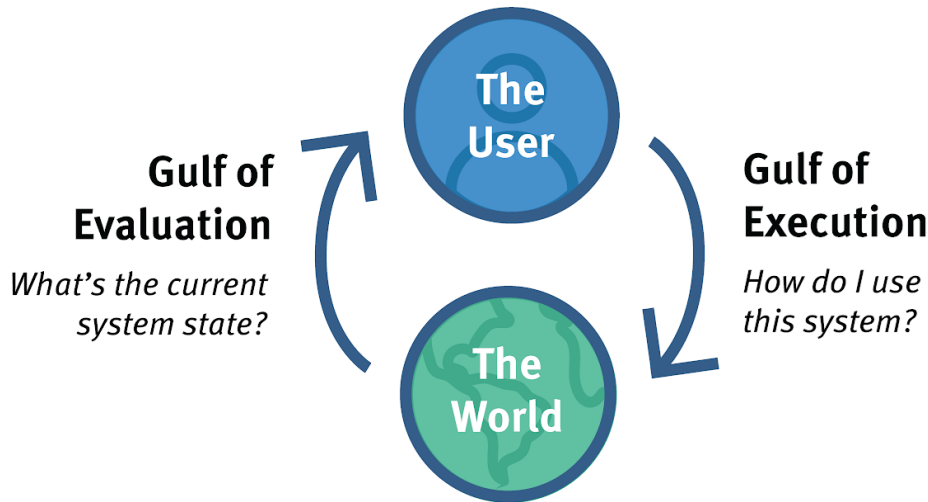
Sign up

Ⓟ Here's what's trending

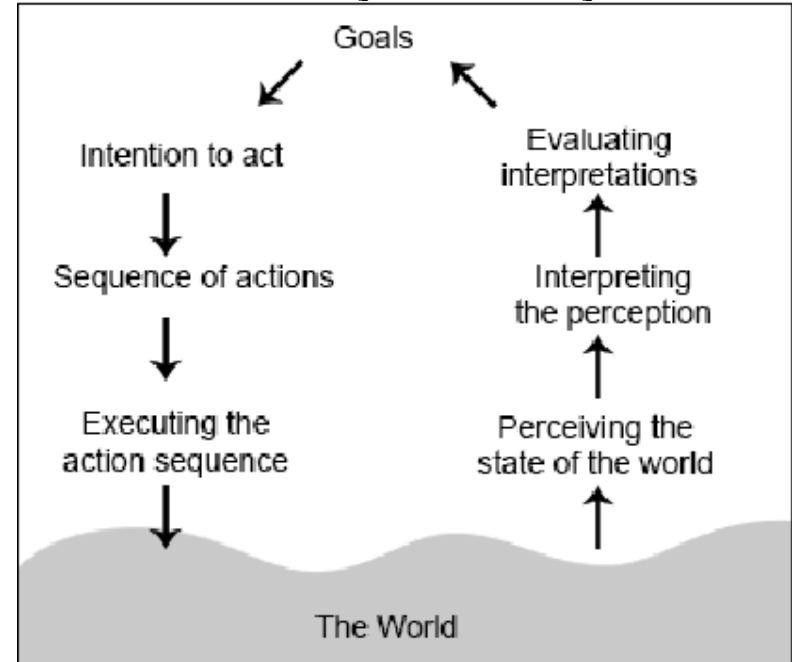
What is Tumblr?

Make things visible: bridge the gulfs of execution and evaluation

As indicated by the figure, users are stuck in an (almost) endless cycle, alternating these two issues: after you execute, you must evaluate the outcome, plan to execute your next step, evaluate that outcome, and so on, until you reach your



nngroup.com **NN/g**



Bluetooth & other devices



Add Bluetooth or other device

Bluetooth



Off

Mouse, keyboard, & pen



Dell KB216 Wired Keyboard

Bluetooth & other devices



Add Bluetooth or other device

Bluetooth



On

Now discoverable as "DESKTOP-ISFRCL"

Mouse, keyboard, & pen



Dell KB216 Wired Keyboard

AUTOMATIC DOWNLOADS



Music



Apps



Books & Audiobooks

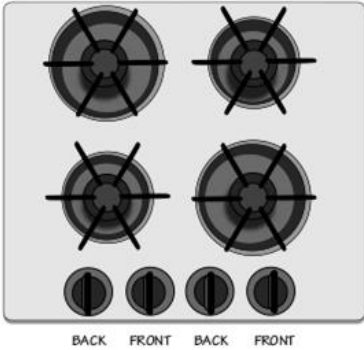


Updates

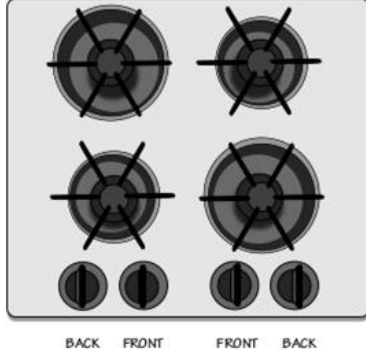


Get the mappings right

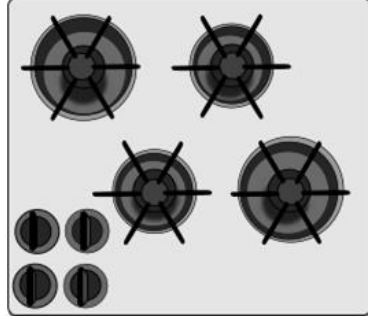
A.



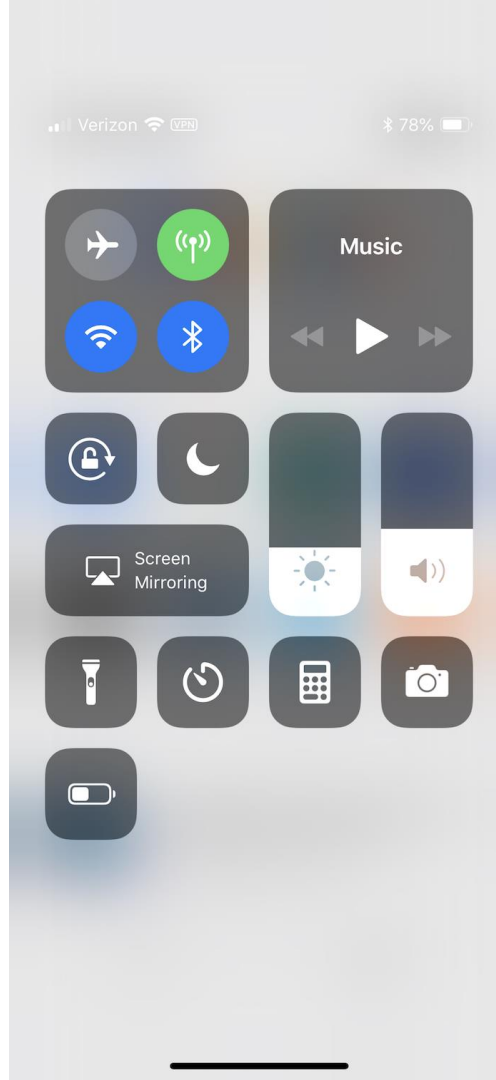
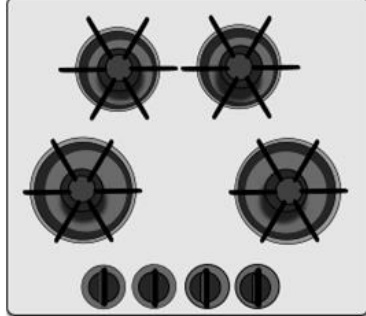
B.



C.

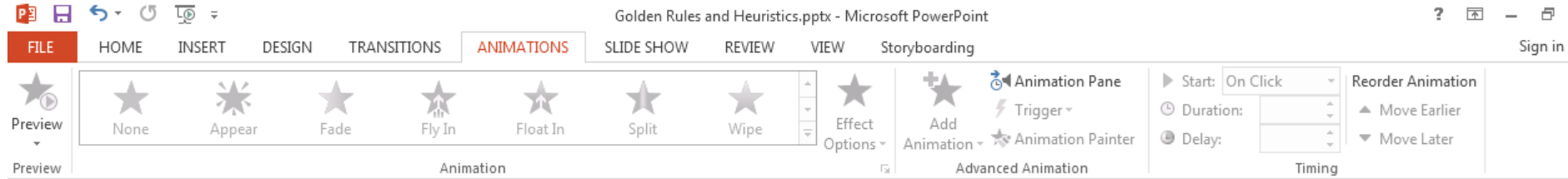


D.



So, put Caps Lock light where Caps Lock button is!

Exploit the power of constraints



Vs.

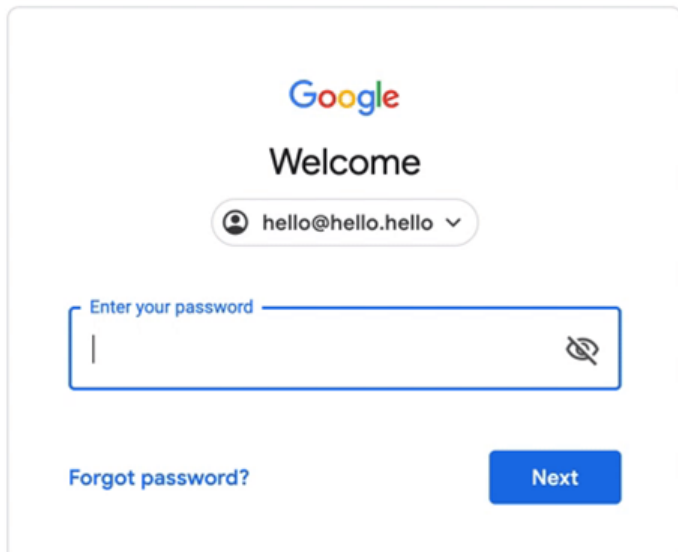


and



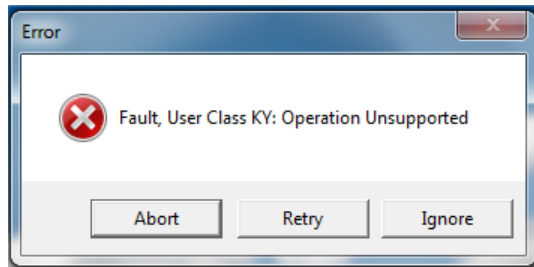
Design for error

If an error is avoidable, it should be dealt with immediately:



A screenshot of a Google login form. At the top is the Google logo, followed by the word "Welcome". Below this is a rounded rectangle containing a user icon, the email address "hello@hello.hello", and a dropdown arrow. Underneath is a password input field with the placeholder text "Enter your password", a vertical cursor, and a toggle icon for visibility. At the bottom left is a link "Forgot password?" and at the bottom right is a blue "Next" button.

Bad error handling:



When all else fails, standardize

